

David Duke

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Summary

Full-stack engineer with over 10 years of experience in full-stack web development. Proficient in frontend technologies (**React, Vue, and Angular**) and backend technologies (**Python, Node.js, Java, C#, PHP, Ruby on Rails**) with a focus on delivering high-quality, scalable solutions. Adept at leveraging cloud services and implementing modern web technologies to enhance application performance and user experience

Experience

UnitedHealth Group

Senior Full Stack Engineer | 01/2020 - Present

Skills: JavaScript, TypeScript, React.js, Redux, Next.js, Node.js, Express.js, Python, Django, FastAPI, Flask, Django Rest Framework, Pandas, Numpy, Pytorch, Tensorflow, GraphQL, Redis, Elasticsearch, Kafka, RabbitMQ, Kubernetes, Docker, API Gateway, AWS(S3, Lambda, EC2, DynamoDB, RDS, EKS, CloudWatch), Jest, JUnit, Mockito, Cypress, WebSocket, OAuth2, JWT, RBAC, CQRS, Prometheus, Grafana, PostgreSQL, MongoDB

- Developed the **COBALT Mental Health Platform** which provided healthcare workers, especially during stressful times like the COVID-19 pandemic, with easy access to mental health support.
- Built a responsive, user-friendly interface using **React**, and **TypeScript** to deliver seamless user experiences.
- Built and maintained backend microservices with **Node.js**, focusing on data processing and secure storage for healthcare applications and **Python** to handle data analytics and reports and storage securely.
- Used asynchronous communication between the microservices via **RabbitMQ** or **Kafka**.
- Implemented **OAuth** and **JWT(JSON Web Token)** for secure user authentication, meeting stringent healthcare industry standards for data privacy and security.
- Used **PostgreSQL**(relational Database) for structured data and implemented **data partitioning** and **sharding** to maintain low latency queries and data availability and **MongoDB** for unstructured, rapidly changing data.
- Managed cloud infrastructure on **AWS** leveraging **API Gateway** for efficient API management, **Lambda** for serverless functions, **EC2** for computing, **S3** for secure video storage.
- Utilized **Docker** for containerization and orchestration tools (**Kubernetes**), and **Jenkins** for CI/CD, ensuring efficient and reliable deployments.
- Conducted **unit testing** with **Jest**, **JUnit**, **Mockito** and **React Testing Library**, improving system reliability.
- Automated the testing pipeline with CI/CD tools (**Github Actions**, **Jenkins**, **CircleCI**), running tests on every commit or pull request.
- Worked closely with cross-functional teams in an **Agile** environment, contributing to sprint planning, code reviews, and continuous integration/continuous delivery (**CI/CD**) pipelines to maintain high code quality and rapid release cycles.

Binance

Software Engineer | 03/2017 - 12/2019

Skills: C++, Java, Spring, Spring Boot, C#, .Net, Perl, PHP, Laravel, Ruby on Rails, JavaScript(ES6+), TypeScript, Node.js, CSS, SASS, LESS, SQL, React, Angular, Redux, AWS(EC2, S3, Lambda, RDS, CloudWatch), Docker, Kubernetes, Jenkins, RabbitMQ, Kafka, JUnit, Mockito, Postman, PostgreSQL, MongoDB, MySQL, Redis, Jenkins, Git, Hibernate, Python(FastAPI, Django, Pytorch, TensorFlow),

- Developed and scaled the **core cryptocurrency exchange platform**, handling millions of trades per second with high availability and low-latency execution.
- Developed the **Fix** engine and **V2** protocol with **Perl**, **Bash** or **PHP** scripts and built several internal services using **Java** and **C#**.
- Optimized backend microservices using **Node.js**, **Ruby on Rails**, **PHP/Symfony**, enabling seamless trade execution and low-latency interactions and integrated the AI model using **Python**(Django).
- Collaborated with frontend teams to integrate **React**, **Angular**-based UIs, ensuring responsive and efficient real-time user interactions, and used various **CSS** preprocessors to make their code look elegant and stylish: **SASS**, **LESS**, **Stylus** with a tiny bit of writing done in pure CSS.
- Designed and implemented real-time trading engines that processed buy and sell orders, ensuring accurate order matching and timely trade execution. Worked extensively on **WebSocket** and **REST APIs** for real-time data exchange between clients and servers, enhancing user experience by delivering live market updates.
- Integrated security measures such as encryption and two-factor authentication (2FA), ensuring the platform's data integrity and user safety.

- Managed and optimized **MySQL**, deployed on **AWS RDS** to store large volumes of transaction history, market data, and user profiles with a focus on performance and reliability.
- Implemented liquidity management systems to ensure smooth trading across the platform, maintaining optimal market conditions for users.

Shopify

Software Engineer | 05/2015 - 03/2017

Skills: Java(Core Java, Java EE, Java Servlets), Node.js, Express.js, JavaScript, React, HTML5, CSS3, jQuery, AJAX, SQL(Oracle DB), MogoDB, JUnit, Mockito, Jenkins, Kubernetes, GitLab, WebSocket, OAuth, JWT, GraphQL,

- Developed and maintained a responsive, user-friendly frontend for Shopify POS using **React** and **TypeScript**, improving the user experience and increasing usability for retail merchants.
- Developed secure, efficient server-side applications with **Java**, **Spring Framework** and **Node.js** and implemented real-time order processing and payment confirmation, using **WebSocket** for bidirectional communication between the client and server, keeping users updated with instant transaction statuses.
- Implemented real-time data synchronization between the Shopify e-commerce platform and the physical POS system, leveraging **RestAPI**, **GraphQL**, and **WebSockets** to ensure smooth, continuous operation.
- Implemented secure authentication using **OAuth** and **JWT** for POS login and user role management, enhancing data privacy and access control.
- Optimized and maintained relational database schemas in **Oracle DB**, ensuring efficient storage and retrieval of high-volume **patient information**.
- Deployed the application on **Azure**, using **Kubernetes** for orchestration and **Gitlab** for CI/CD, reducing deployment times by 40%.
- Collaborated in the integration of new features for inventory management, custom pricing, and customer loyalty programs, enhancing the flexibility of the POS system for various retail needs.

Skills

Node.js, Nest.js, Express.js, JavaScript, TypeScript, Python, Django, Flask, FastAPI, Numpy, Pandas Java, Spring, Spring Boot, Go, Gin, React, Angular, Vue, Next.js, Redux, HTML 5, CSS3, Bootstrap, MySQL, PostgreSQL, MongoDB, Redis, Docker, Kubernetes, Jenkins, Git Lab, Github Action, AWS, Lambda, EC2, S3, API Gateway, Azure, CI/CD, RESTful API, GraphQL, Microservices Architecture, Microfrontend, Scalable Web Application, Git, Agile/Scrum, Kanban, PHP, Perl, Ruby on Rails

Education

University of Kentucky
Computer Science | 05/2015